

WASP-10b

R-0862

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_240905 · created 2026-07-28T22:19:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460558.7694 +/- 0.0054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.037
Transit depth (Rp/Rs)^2	2.11 +/- 1.08 [%]
Orbital Inclination (inc)	86.94 +/- 1.41
Airmass coefficient 1 (a1)	0.969 +/- 0.016
Airmass coefficient 2 (a2)	0.027 +/- 0.064
Scatter in the residuals of the lightcurve fit is	1.71 %
Best Comparison Star	#1 - [211, 415]
Optimal Method	PSF photometry
Transit Duration (day)	0.074 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.145 ± 0.037 vs 0.1592 (deviation 0.38σ)
Duration fit vs published	0.074 d vs 0.0946 d (deviation 1.03σ)
Mid-time O-C	-11.7 min
Depth SNR	3.9
Residual scatter	1.71 %

Light curve

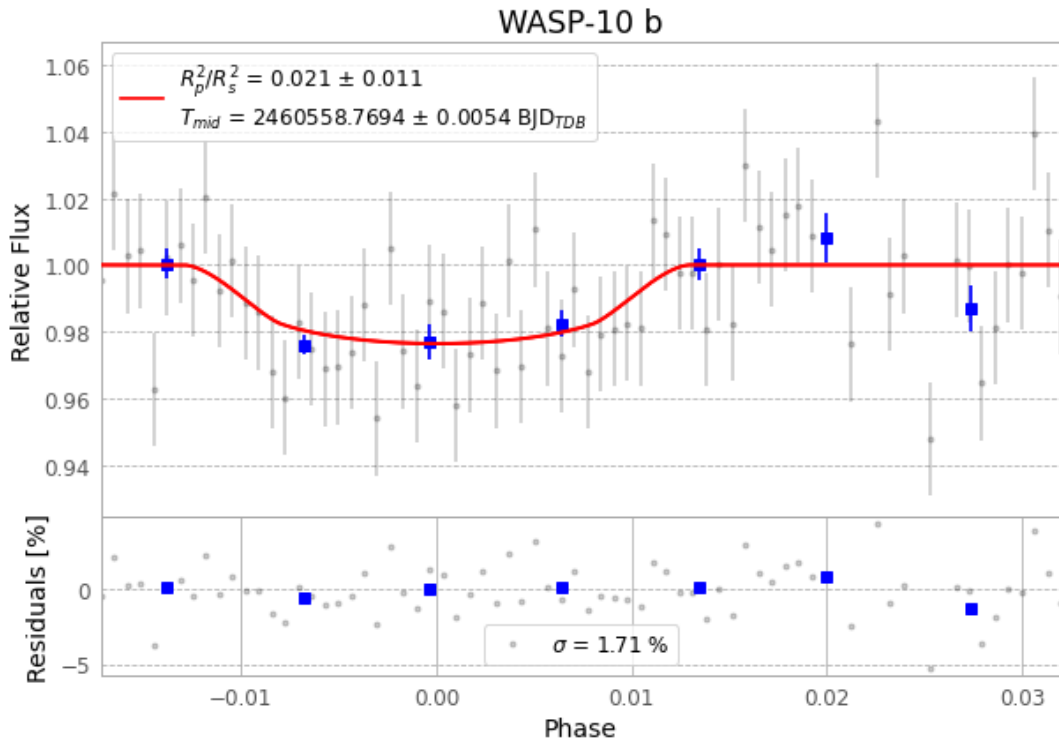


Figure 1 — FinalLightCurve_WASP-10 b_2024-09-04.png (SHA-256 ea4f4fe0292e3169...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [335, 195], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b4dc19e5819f855a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

74 FITS frames (49 MB), WASP-10240905050316.FITS → WASP-10240905084215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-240905.log (cec0e87e82f4...), FinalLightCurve_WASP-10 b_2024-09-04.png (ea4f4fe0292e...), FinalLightCurve_WASP-10 b_2024-09-04.csv (486bd9a066c3...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 22:19Z.